Extreme Values for Multivariable Functions
Lecture 5 for Multivariable Calculus

Contents

1 Lecture Scope and Goals

In this lecture, we focus on extreme values of multivariable functions. The goals are:

- distinguish local extrema and global extrema,
- understand when global extrema must exist (compactness),
- find interior candidates using first-order conditions,

- classify interior critical points using second derivatives,
- solve absolute-extrema problems on closed and bounded regions,
- use Lagrange multipliers for smooth constraints.

Central Theme

For multivariable optimization, the full answer usually comes from *three places*: interior critical points, boundary points, and corner/end points.

2 Local vs Global Extrema

Let $f : D \subset \mathbb{R}^2 \to \mathbb{R}$.

2.1 Definitions

- f has a **local maximum** at $p \in D$ if there exists $r > 0$ such that
$$f(p) \geq f(x,y) \quad \text{for all } (x,y) \in D \cap B_r(p).$$
- f has a **local minimum** at p if
$$f(p) \leq f(x,y) \quad \text{for all } (x,y) \in D \cap B_r(p).$$
- f has an **absolute (global) maximum** at p if
$$f(p) \geq f(x,y) \quad \text{for all } (x,y) \in D.$$
- f has an **absolute (global) minimum** at p if
$$f(p) \leq f(x,y) \quad \text{for all } (x,y) \in D.$$

Example 1. For $f(x,y) = x^2+y^2$ on $\mathbb{R}^2$, $(0,0)$ is a local minimum and also an absolute minimum. There is no absolute maximum on $\mathbb{R}^2$.

2.2 Necessary Condition at Interior Local Extremum

Theorem 2.1 (Necessary condition for interior local extremum)**.** *Let $D \subset \mathbb{R}^2$, let $p = (a,b)$ be an interior point of D, and let $f : D \to \mathbb{R}$ be differentiable at p. If f has a local maximum or local minimum at p, then*
$$f_x(a,b) = 0, \qquad f_y(a,b) = 0.$$
Equivalently, $\nabla f(a,b) = 0$.

Proof. Assume f has a local extremum at $p = (a,b)$ and is differentiable at p. Since p is an interior point of D, there exists $r > 0$ such that
$$B_r(p) \subset D.$$

First, keep $y = b$ fixed and define

$$\phi(x) = f(x, b), \qquad x \in (a - r, a + r).$$

Because f has a local extremum at (a, b), the one-variable function ϕ has a local extremum at $x = a$. Also, differentiability of f at (a, b) implies the partial derivative $f_x(a, b)$ exists, and

$$\phi'(a) = f_x(a, b).$$

By Fermat's theorem in one variable, $\phi'(a) = 0$, hence $f_x(a, b) = 0$.

Next, keep $x = a$ fixed and define

$$\psi(y) = f(a, y), \qquad y \in (b - r, b + r).$$

Similarly, ψ has a local extremum at $y = b$, and

$$\psi'(b) = f_y(a, b).$$

Again by Fermat's theorem, $\psi'(b) = 0$, so $f_y(a, b) = 0$.

Therefore $f_x(a, b) = f_y(a, b) = 0$, i.e. $\nabla f(a, b) = 0$. $\square$

Critical point terminology. A point is called **critical** if $\nabla f = 0$ or one of the first partial derivatives does not exist.

3 Existence of Absolute Extrema

3.1 Extreme Value Theorem (EVT)

If $D \subset \mathbb{R}^2$ is *closed and bounded* and f is continuous on D, then f attains both an absolute maximum and an absolute minimum on D.

Why this matters. In many optimization problems, you are guaranteed that a best and worst value exist before calculating anything.

Example 2 (existence guaranteed). Let

$$f(x, y) = x^2 + 2y^2 - 2x + 1$$

on the disk $x^2 + y^2 \leq 4$. The region is closed and bounded, and f is continuous, so absolute max/min exist.

Example 3 (no guarantee). Take $f(x, y) = x^2 + y^2$ on the open disk $x^2 + y^2 < 1$. No absolute maximum exists, because values approach 1 near boundary but boundary points are not included.

4 Second Derivative Test in Two Variables

Theorem 4.1 (Second derivative test in $\mathbb{R}^2$). *Let $f \in C^2$ in a neighborhood of (a,b), and suppose (a,b) is a critical point:*

$$f_x(a,b) = f_y(a,b) = 0.$$

Set

$$A = f_{xx}(a,b), \quad B = f_{xy}(a,b), \quad C = f_{yy}(a,b), \quad D = AC - B^2.$$

Then:

- *If $D > 0$ and $A > 0$, then (a,b) is a strict local minimum.*
- *If $D > 0$ and $A < 0$, then (a,b) is a strict local maximum.*
- *If $D < 0$, then (a,b) is a saddle point.*
- *If $D = 0$, this test is inconclusive.*

Proof. Write

$$h = x - a, \qquad k = y - b.$$

Since $f \in C^2$, Taylor expansion at (a,b) gives

$$f(a+h, b+k) - f(a,b) = \frac{1}{2}\big(Ah^2 + 2Bhk + Ck^2\big) + r(h,k),$$

where

$$\frac{r(h,k)}{h^2+k^2} \to 0 \quad \text{as } (h,k) \to (0,0).$$

So the local sign is determined by the quadratic form

$$Q(h,k) = Ah^2 + 2Bhk + Ck^2.$$

Case 1: $D > 0$. Then

$$Q(h,k) = A\left(h + \frac{B}{A}k\right)^2 + \frac{D}{A}k^2$$

(here $A \neq 0$ because $D > 0$). If $A > 0$, both coefficients are positive, so $Q(h,k) > 0$ for $(h,k) \neq (0,0)$. Hence for sufficiently small $(h,k) \neq (0,0)$, the remainder cannot change the sign, and

$$f(a+h, b+k) - f(a,b) > 0.$$

Thus (a,b) is a strict local minimum.

If $A < 0$, then both coefficients are negative, so $Q(h,k) < 0$ for $(h,k) \neq (0,0)$. Similarly,

$$f(a+h, b+k) - f(a,b) < 0$$

for all sufficiently small nonzero (h,k), so (a,b) is a strict local maximum.

Case 2: $D < 0$. Consider

$$q(t) = Q(1,t) = A + 2Bt + Ct^2.$$

Its discriminant is

$$\Delta = (2B)^2 - 4AC = 4(B^2 - AC) = -4D > 0,$$

so q has two distinct real roots and therefore takes both positive and negative values. Hence there exist directions $(1, t_1)$ and $(1, t_2)$ such that

$$Q(1, t_1) > 0, \qquad Q(1, t_2) < 0.$$

Along the rays $(h, k) = s(1, t_i)$ with $s \to 0$, the leading term has opposite signs, and the remainder is $o(s^2)$, so

$$f(a + h, b + k) - f(a, b)$$

is positive in one direction and negative in another. Therefore (a, b) is a saddle point.

Case 3: $D = 0$. The quadratic part is semidefinite or degenerate, so higher-order terms may decide the type; no general conclusion follows. □

Pitfall 1

The second derivative test classifies *interior critical points only.* It does not replace boundary analysis for absolute extrema on a region.

Example 4. Let

$$f(x, y) = x^3 - 3xy + y^3.$$

First derivatives:

$$f_x = 3x^2 - 3y, \qquad f_y = -3x + 3y^2.$$

Solve

$$x^2 = y, \qquad y^2 = x.$$

Critical points are $(0, 0)$ and $(1, 1)$.

Second derivatives:

$$f_{xx} = 6x, \quad f_{yy} = 6y, \quad f_{xy} = -3.$$

- At $(0, 0)$: $D = 0 \cdot 0 - (-3)^2 = -9 < 0$, so saddle point.
- At $(1, 1)$: $D = 6 \cdot 6 - (-3)^2 = 27 > 0$, and $f_{xx}(1, 1) = 6 > 0$, so local minimum.

5 Absolute Extrema on Closed Regions

5.1 Standard Workflow

For continuous f on a closed bounded region D:

- Find interior critical points ($\nabla f = 0$) inside D.
- Analyze each smooth boundary piece (parameterization or substitution to one-variable problem).
- Check corner/end points.
- Compare all candidate values; largest is absolute max, smallest is absolute min.

Example 5 (rectangle). Find absolute extrema of

$$f(x,y) = x^2 + y^2 - 2x - 4y$$

on

$$D = [0,3] \times [0,4].$$

Interior.

$$f_x = 2x - 2, \qquad f_y = 2y - 4.$$

Critical point: $(1,2)$ (inside D), and

$$f(1,2) = 1 + 4 - 2 - 8 = -5.$$

Boundary $x = 0$, $0 \le y \le 4$.

$$f(0,y) = y^2 - 4y, \quad (f(0,y))' = 2y - 4 = 0 \Rightarrow y = 2.$$

Values:

$$f(0,0) = 0, \quad f(0,2) = -4, \quad f(0,4) = 0.$$

Boundary $x = 3$, $0 \le y \le 4$.

$$f(3,y) = y^2 - 4y + 3, \quad (f(3,y))' = 2y - 4 = 0 \Rightarrow y = 2.$$

Values:

$$f(3,0) = 3, \quad f(3,2) = -1, \quad f(3,4) = 3.$$

Boundary $y = 0$, $0 \le x \le 3$.

$$f(x,0) = x^2 - 2x, \quad (f(x,0))' = 2x - 2 = 0 \Rightarrow x = 1.$$

Values:

$$f(0,0) = 0, \quad f(1,0) = -1, \quad f(3,0) = 3.$$

Boundary $y = 4$, $0 \le x \le 3$.

$$f(x,4) = x^2 - 2x, \quad (f(x,4))' = 2x - 2 = 0 \Rightarrow x = 1.$$

Values:

$$f(0,4) = 0, \quad f(1,4) = -1, \quad f(3,4) = 3.$$

Compare all candidates:

$$\min f = -5 \text{ at } (1,2), \qquad \max f = 3 \text{ at } (3,0) \text{ and } (3,4).$$

6 Constrained Extrema and Lagrange Multipliers

6.1 Core Condition

To optimize $f(x,y)$ subject to $g(x,y) = c$ (with $\nabla g \neq 0$ at candidates), solve:

$$\nabla f = \lambda \nabla g, \qquad g(x,y) = c.$$

6.2 Geometric Idea

Suppose we optimize $f(x, y)$ on the constraint curve

$$g(x, y) = c.$$

There are several equivalent geometric views:

1) Level-curve tangency picture. Think of level curves $f(x, y) = k$ as a moving family. As k increases (or decreases), these curves slide. At the constrained optimum, the highest/lowest level curve that still touches $g = c$ will typically *just touch* it (tangency), rather than cross it. So near that point, the two curves share the same tangent direction.

2) Tangent-direction derivative picture. Let p be a constrained extremum and let v be any tangent vector to the constraint at p. Moving along the constraint in direction v is the only feasible first-order motion, so the first-order change must vanish:

$$D_v f(p) = \nabla f(p) \cdot v = 0.$$

Thus $\nabla f(p)$ is orthogonal to every tangent direction of the constraint.

3) Why this gives $\nabla f = \lambda \nabla g$. For a regular point of the constraint ($\nabla g(p) \neq 0$), $\nabla g(p)$ is normal to the curve $g = c$. From the previous paragraph, $\nabla f(p)$ is also normal to the same curve. In $\mathbb{R}^2$, two nonzero normals to the same tangent line are parallel, so

$$\nabla f(p) = \lambda \nabla g(p)$$

for some scalar λ.

4) Higher-dimensional intuition. For constraints in $\mathbb{R}^n$, feasible directions form the tangent space of the constraint surface. At constrained extrema, ∇f is orthogonal to that tangent space, hence it lies in the normal space generated by constraint gradients. This is the geometric core behind all Lagrange multiplier formulas.

Important note. The equation $\nabla f = \lambda \nabla g$ gives *necessary* candidates. You still need to evaluate f at all candidates (and possibly endpoints/corners for non-smooth constraints) to decide max/min.

Example 6. Find constrained extrema of

$$f(x, y) = xy, \qquad x^2 + y^2 = 1.$$

Set

$$\nabla f = (y, x), \qquad \nabla g = (2x, 2y),$$

so

$$y = 2\lambda x, \qquad x = 2\lambda y, \qquad x^2 + y^2 = 1.$$

From the first two equations, $x = \pm y$.

- If $x = y$, then $2x^2 = 1 \Rightarrow x = \pm\frac{1}{\sqrt{2}}$ and

$$f = xy = \frac{1}{2}.$$

- If $x = -y$, then

$$f = xy = -\frac{1}{2}.$$

Therefore

$$f_{\max} = \frac{1}{2}, \qquad f_{\min} = -\frac{1}{2}.$$

Pitfall 2

Lagrange multipliers provide candidates on smooth constraints. Always check all candidate points and verify the constraint is satisfied.

7 Example Bank (Board-Ready)

7.1 Example 7 (Quick Classification)

Classify the critical point(s) of

$$f(x, y) = x^2 - y^2 + 4x - 2y.$$

Solution.

$$f_x = 2x + 4, \qquad f_y = -2y - 2.$$

Critical point: $(-2, -1)$. Second derivatives:

$$f_{xx} = 2, \quad f_{yy} = -2, \quad f_{xy} = 0,$$

so

$$D = 2(-2) - 0 = -4 < 0.$$

Hence $(-2, -1)$ is a saddle point.

7.2 Example 8 (Absolute Extrema on a Disk)

Find absolute extrema of

$$f(x, y) = x^2 + y^2 - 2x$$

on $x^2 + y^2 \le 4$.

Solution. Interior:

$$f_x = 2x - 2, \quad f_y = 2y \Rightarrow (1, 0),$$

with $f(1, 0) = -1$.

Boundary $x^2 + y^2 = 4$: write

$$f = 4 - 2x.$$

So on boundary, max occurs when $x = -2$, min occurs when $x = 2$:

$$f(-2, 0) = 8, \qquad f(2, 0) = 0.$$

Compare with interior value -1:

$$f_{\min} = -1 \text{ at } (1, 0), \qquad f_{\max} = 8 \text{ at } (-2, 0).$$

7.3 Example 9 (Constrained Extremum)

Find max/min of

$$f(x, y) = x + y$$

subject to

$$x^2 + 4y^2 = 4.$$

Solution.

$$\nabla f = (1, 1), \quad \nabla g = (2x, 8y), \quad (1, 1) = \lambda(2x, 8y).$$

Hence

$$x = \frac{1}{2\lambda}, \qquad y = \frac{1}{8\lambda}.$$

Substitute into constraint:

$$\frac{1}{4\lambda^2} + 4 \cdot \frac{1}{64\lambda^2} = 4 \Rightarrow \frac{5}{16\lambda^2} = 4 \Rightarrow \lambda^2 = \frac{5}{64}.$$

So $\lambda = \pm\frac{\sqrt{5}}{8}$, giving points

$$\left(\frac{4}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right), \quad \left(-\frac{4}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\right).$$

Function values:

$$f = \pm\sqrt{5}.$$

Therefore absolute maximum is $\sqrt{5}$, absolute minimum is $-\sqrt{5}$.

8 Mini Workflow for Students

For a typical exam optimization problem:

1. Identify domain type: unconstrained, closed region, or constrained curve/surface.
2. Write all candidate equations carefully.
3. Solve algebra system cleanly.
4. Evaluate f at every candidate.
5. State final answer with points and values.

9 Practice Set (In-Class or Homework)

1. Find and classify all critical points of

$$f(x, y) = x^4 + y^4 - 4x^2 - 4y^2.$$

2. Find absolute extrema of

$$f(x, y) = x^2 + y^2 + 2x - 4y$$

on triangle with vertices $(0, 0), (2, 0), (0, 2)$.

3. Use Lagrange multipliers to optimize

$$f(x, y) = x^2 + y^2$$

subject to

$$x + 2y = 6.$$

4. Determine whether $f(x, y) = x^2 + y^2$ has an absolute maximum on

$$D = \{(x, y) : x^2 + y^2 < 9\},$$

and explain.

5. Let

$$f(x, y) = x^3 + y^3 - 3xy.$$

Find all critical points and classify them.

Solution Sketches

1. Critical points are $(0, 0), (\pm\sqrt{2}, 0), (0, \pm\sqrt{2}), (\pm\sqrt{2}, \pm\sqrt{2})$ by solving

$$4x(x^2 - 2) = 0, \quad 4y(y^2 - 2) = 0.$$

Use Hessian to classify: $(0, 0)$ local max, $(\pm\sqrt{2}, \pm\sqrt{2})$ local mins, mixed-sign points are saddles.

2. Interior critical point $(-1, 2)$ is outside triangle; only boundary matters. Check three edges as one-variable functions and compare endpoint/interior-edge values.

3. Constraint gives $x = 6 - 2y$, so

$$f(y) = (6 - 2y)^2 + y^2 = 5y^2 - 24y + 36.$$

Minimum at $y = \frac{12}{5}$, $x = \frac{6}{5}$. No maximum on this line.

4. No absolute maximum on open disk: supremum is 9 but not attained. Absolute minimum is 0 at $(0, 0)$.

5. Same critical equations as Example 4:

$$x^2 = y, \quad y^2 = x.$$

Points $(0, 0)$ and $(1, 1)$. $(0, 0)$ saddle, $(1, 1)$ local minimum.

10 Summary

- Interior local extrema require $\nabla f = 0$ (or non-differentiability).
- Use Hessian test to classify interior critical points when $D \neq 0$.
- For absolute extrema on closed bounded regions: interior + boundary + corners.
- For smooth equality constraints: use Lagrange multipliers.
- Final answers must include both points and function values.